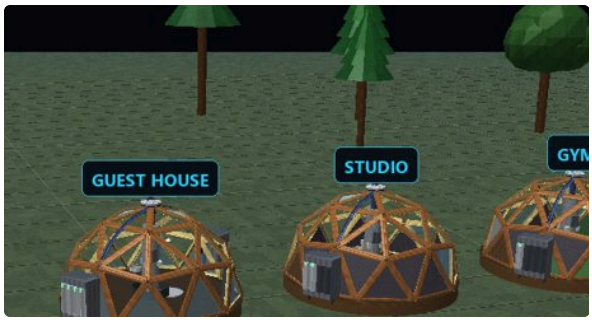
MODULAR HOUSING · BY A VETERAN, FOR VETERANS

The 40 Hour Cabin

One standard body, made the same way every time.

A small home made like a product: one standard frame, a shell in 4 slices, a utility core that carries every service, and snap-in panels that decide what the building is. The standard building lists at $12,036, and every figure on this page is computed by open-source software anyone can run.



$43.45
a sq ft for the standard building, margin in. Published: new house $217, manufactured $87
MODELLED

$1,890
utility core: 5 services, a 25-year life, moves to the next dome for $260
MODELLED

9,870 gal
of rain a year off its own seams, at 40 in of rain
MODELLED

12 + 25
snap-in panel types and modules: a bay changes with a driver, not a builder
MODELLED

15
buildings from one body, $12,036 to $25,511
MODELLED

32,882
veterans homeless on one night in January 2024; 13,851 unsheltered
PUBLISHED (HUD)

THE PROBLEM

Shelter eats the time and money a family needs for everything else: a new site-built house costs **$217 a square foot** before the land, a new manufactured home **$87** (NAHB, Census). On one night in January 2024, **32,882 veterans** had no home of their own (HUD).

THE IDEA: A HOME MADE LIKE A PRODUCT

A manufactured home built as a system, not a one-off. The frame is 40 flat panels on two strut lengths, so every size from 10 to 30 feet uses the same 120 members, the same jig and the same hardware set. The shell comes in 4 slices that two people carry and snap together. Everything that makes it a particular building plugs in.

WHAT PLUGS IN

- **A utility core.** One column carries air, drain, power, structure, water: $1,890, 30% of the build, a 25-year life. Buy the plumbing once.
- **Rain, built in.** The seams between panels are channels: 247 gallons for every inch that falls on 550 sq ft of shell, run to the pad's tank. No gutters.
- **Snap-in panels.** Windows, doors, skylights, solar bays, flues: 12 types and 25 modules, $180 to $1,480.
- **A skin you renew.** Two layers of all-cotton duck soaked in silicone, $1,486 against $2,809 in sailcloth, and a beeswax coat every 1–2 years, about $34 a coat (estimated).

Stem cell · Gym · Studio · Nursery · Guest house · Workshop · Garage · Home · Sauna · Food · Advertiser · Cold storage · Bunker · Treehouse · Jacuzzi: one body, 15 buildings. It stands on a serviced pad and can move: **own the building, rent the ground**.

WHO

Donovan Zeanah. U.S. Navy veteran: six years as a Mass Communication Specialist, three of them in Japan. Learned to program and was certified as an avionics bench technician; now studying for an A&P certificate on the G.I. Bill, on the way to aerospace engineering. Built so far, in the open: the solver, two books (83 and 55 chapters), 34 films and a 6-rung family edition.

WHY I AM DOING THIS

> *Buckminster Fuller wanted housing cheap enough that a mother is not so burdened by it she cannot raise her child — what I took from him, not his words. I grew up with a single parent and a repossessed trailer, and came home from the Navy asking what a small home I owned would have done for my eighteen-year-old self.*

BUILT BY A VETERAN, FOR VETERANS

The life this fits is one a lot of us already know: a small, squared-away space; a system you maintain on a schedule; parts you swap in the field; a crew that builds together. A village of 49 standard buildings on pads, the size of the Veterans Community Project's Kansas City village (VCP), is $0.90 million in buildings and pads before land. It is designed to sit beside HUD-VASH, SSVF, VA Grant and Per Diem, and to be built and kept by veterans.

MEASURED, MODELLED, NOT YET

MEASURED	splitting, 30 wedges per 4-hour session; strut weights; a full-size prototype frame.
MODELLED	cost, energy, rain and the core, in open code.
NOT YET	an engineer's rating; an instrumented year ($12,850); the moulded shell and strut tooling ($60,800).
AGAINST US	53% of the floor has 6.67 ft of headroom; the whole building is about 67 hours, not forty.

THE ASK: $100,650

The sum of its lines, not a round number: prove it, then make it repeatable.

Prove it — A full test platform and one standing dome · Sensors in the wall · An engineer	**$12,850**
Make it repeatable — CNC router · CNC plasma table · Laser cutter · Tooling to mould a strut with its hardware in it · Automated processing for the repetitive work	**$60,800**
Build what an owner cannot — Utility hub · Composite triangle	**$22,000**
Seed the network — Seeding the quilt network	**$5,000**
Total	**$100,650**

Costed as the next step, not asked here: a first site of 3 serviced pads, $100,000.

Every figure computed 13 October 2026 by the project's own software and labelled by kind. Code, books and films: **github.com/dkzeanah/DomeSimulator**

zeanahlab.com · Instagram @DonovanZeanah · TikTok @shortcircuitr5 · Facebook facebook.com/zeanah · GitHub github.com/dkzeanah

The The 40 Hour Cabin: the numbers

What one building costs, what it sells for, what the money buys, and what argues back.

ONE STANDARD BUILDING: COST TO PRICE

MODELLED

Wedge-cut frame	$692
Shower-cap shell	$3,265
Utility column and seal cap	$1,405
Light, fan and cooling	$485
Utility panels	$491
Build labour	$1,884
Materials and labour	$8,221
Overhead	$1,480
Warranty reserve	$329
Cost to build	**$10,030**
Maker's margin, 20% on cost (declared)	$2,006
List price	**$12,036**

THE SAME 277 SQ FT, BOUGHT THREE WAYS

PUBLISHED · MODELLED

	for the floor	a sq ft
New site-built house, without land (NAHB)	$60,107	$217
New manufactured home (Census 2023)	$23,993	$87
This standard building	$12,036	$43.45
... standing on its pad, at cost	$18,276	$65.98

WHAT THE ASK BUYS: $100,650

THE CAMPAIGN'S COSTED LINES

Prove it	$12,850
Make it repeatable	$60,800
Build what an owner cannot	$22,000
Seed the network	$5,000
Margins that repay the ask, at one building's margin each	51 buildings
Startup model: equipment + working capital	$86,000 + $14,000
Buildings the startup's working capital funds	

FOR THE OWNER

MODELLED · ILLUSTRATIVE

Heating and cooling, a year: this building / a box of the same floor	$129 / $220
Never spent, every year	$91
Utility core, over its 25-year life	$76 a year
Moving the core to the next building	$260
For comparison: a $100,000 mortgage at 6.5% for 30 years	$227,544 repaid, $127,544 of it interest
This building on its pad, at cost, paid once	$18,276

FOR A PAD HOST: A $10,000 PAD (THE AUTHOR'S TARGET) AT 80% OCCUPANCY

SCENARIO

rent	after costs	payback
$500 a month	$3,880 a year net	30.9 months to repay
$1,000 a month	$8,680 a year net	13.8 months to repay
$1,500 a month	$13,480 a year net	8.9 months to repay
Management, tax and insurance, a year	$920	

CAMPAIGN REWARDS: WHAT EACH TIER LEAVES

MODELLED

tier	pledge	cost to send	kept
The drawings and the numbers	$35	$2	94%
Quilter's kit	$95	$58	39%
One bay's hardware set	$180	$112	38%
A shower cap for your dome	$1,450	$980	32%
The utility hub	$2,400	$1,620	32%
Dome kit, frame included	$19,800	$14,900	25%
Dome kit, bring your own trees	$11,400	$8,100	29%
Pad host's pack	$640	$395	38%

FOR SCALE

PUBLISHED · MODELLED

Unsheltered veterans, one night in January 2024 (HUD)	13,851
A standard building on its pad for each, at cost, before land	$253,146,626

THE NUMBERS THAT ARGUE BACK

COMPUTED

The starter pad, computed, against the $10,000 target	$13,148 ($3,148 over)
Labour in the standard building	about 67 hours, not forty
Two wedges planed square keep of the log / a mill's 2x4s	53% / 57%
Rated by an engineer	not yet -- the first line of the ask
Margin, prices, rents	declared or estimated, each named in the code

Every figure computed by the project's own models on the day this page was printed; modelled figures are the software's, published ones are cited (NAHB, Census, HUD), declared ones are named with their reason in the code. github.com/dkzeanah/DomeSimulator

THE 40 HOUR CABIN · MODULAR HOUSING · BY A VETERAN, FOR VETERANS

Efficiency, by the numbers

One standard body against the houses it would replace.

this building · the comparison · variant or upkeep

PRICE A SQUARE FOOT

New site-built house, without land **$217**

New manufactured home **$87**

This standard building **$43.45**

The standard building is 50% of a new manufactured home's price a square foot, margin included and the pad extra. Not like for like: one room with a service core, not a finished house.

MODELLED · PUBLISHED (NAHB, CENSUS)

WHERE THE $12,036 GOES



- Frame **$692** 6%
- Weather shell **$3,265** 27%
- Utility column **$1,405** 12%
- Services **$485** 4%
- Utility panels **$491** 4%
- Build labour **$1,884** 16%
- Overhead and warranty **$1,809** 15%
- Maker's margin **$2,006** 17%

The core is $1,890, 30% of the build. It lasts 25 years and moves to the next dome for $260.

MODELLED

OUTSIDE SKIN, SAME FLOOR

Box house **997 sq ft**

Dome **583 sq ft**

42% less to build, to weatherproof, and to lose heat through.

MODELLED

HEATING AND COOLING, A YEAR

Box house **$220**

Dome **$129**

$91 a year never spent, from the shape alone.

MODELLED

WIND LOAD

Box house, drag **1.05**

Dome, drag **0.42**

60% less push from the same wind: the shape sheds it.

PUBLISHED COEFFICIENTS

RAIN THE SEAMS CATCH, A YEAR

20 in of rain a year **4,935 gal**

40 in of rain a year **9,870 gal**

60 in of rain a year **14,805 gal**

550 sq ft of shell, 72% caught: 247 gallons for every inch that falls, run by the seams into the pad's tank. No gutters.

MODELLED · RAINFALL DECLARED

ONE PARTS LIST, THREE SIZES

10 ft across **79 sq ft**

19.4 ft across **296 sq ft**

30 ft across **707 sq ft**

Every size is the same 120 members and 40 panels on the same two lengths: one jig, one hardware set, one way of working.

MODELLED

THE SKIN, AND KEEPING IT

Woven sailcloth **$2,809**

Cotton duck + silicone **$1,486**

Beeswax recoat **$34**

2 layers, 112 m² of 100% cotton soaked in silicone. A beeswax coat every 1–2 years: 2.2 kg, about 6 hours.

ESTIMATED · SEE THE TABLE

WOOD KEPT FROM THE LOG

Sawn into two-by-fours **45%**

A good mill, mixed sizes **60%**

Split into wedges **88%**

The frame comes from the tree where it stands; the corners of a log are not waste.

SIMULATED · MILL ALLOWANCE THE AUTHOR'S

SNAP-IN, NOT BUILT-IN

Flue panel **$180**

Mirror panel **$275**

Skylight panel **$395**

Open-bay door **$1,480**

12 panel types and 25 snap-in modules, from $180 to $1,480. A bay changes with a driver, not a builder.

MODELLED

THE CORE, OVER ITS LIFE

Build a core **$1,890**

Move it to the next dome **$260**

Each year of its 25 **$76**

Buy the plumbing once. The core carries 5 services (air, drain, power, structure, water) and outlives the shell around it.

MODELLED

HOURS OF WORK

Splitting, measured **16 h**

Frame, mistakes allowed for **37.5 h**

Whole building **67 h**

The frame clock keeps 7.5 hours for getting it wrong. The whole building, counted the same way, is about 67 hours.

MEASURED · ESTIMATED

DECLARED, NOT COMPUTED: THE BEESWAX UPKEEP

wax_g_m2	40 g/m2	estimate	a maintenance coat rubbed into cloth already proofed with silicone: far less than a first waxing of bare canvas
wax_usd_kg	15 $/kg	estimate	beeswax pellets bought by the kilogram
wax_m2_hour	10 m2/hour	estimate	rubbing in and melting off with a heat gun
wax_every_lo	1 years	owner's practice	recoat yearly where it is wet and sunny
wax_every_hi	2 years	owner's practice	every second year where it is not

Every bar drawn to scale from the project's own models on 13 October 2026. Sources and code: **github.com/dkzeanah/DomeSimulator**. Modelled figures are what the software computes; published ones are cited in its source; estimates are declared above or in the module that uses them.